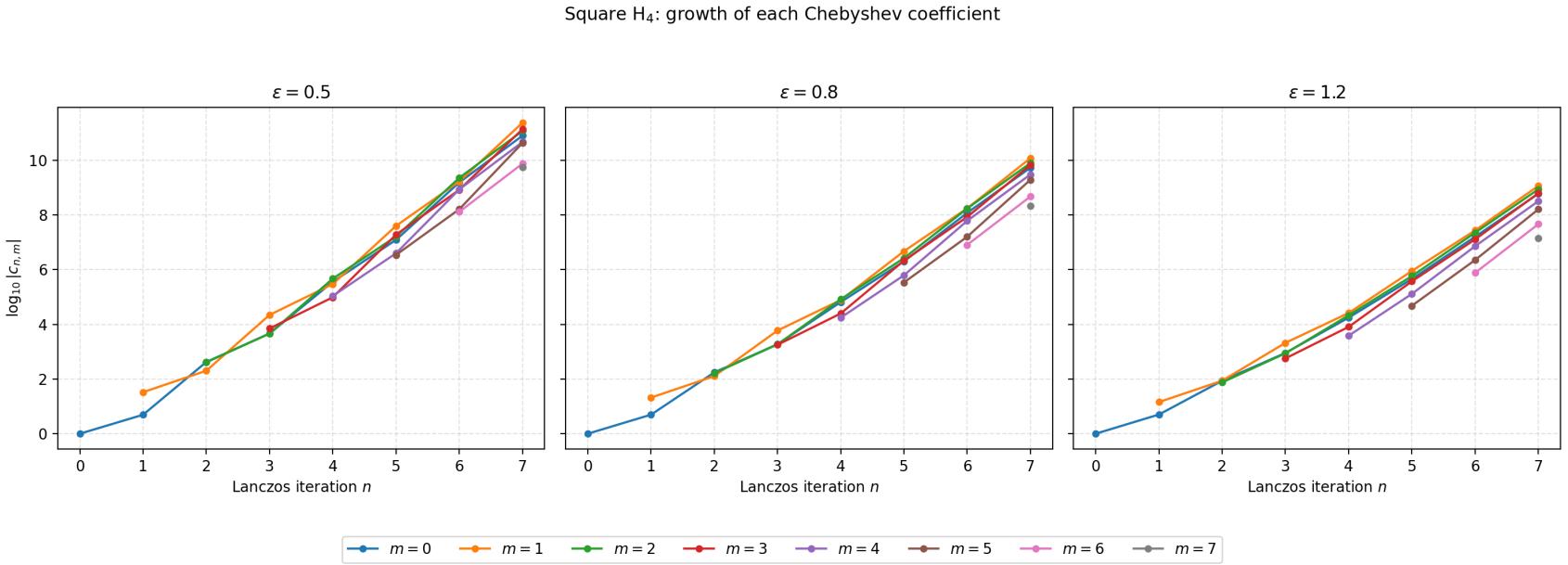
How the OCQKUDSIN Chebyshev coefficients grow

Square H_4; $\epsilon = 0.5,\ 0.8,\ 1.2$; eight Lanczos states

What is plotted. For $H = \sum_\ell h_\ell P_\ell$, these three runs use the sine operator and represent each normalized Lanczos state by

$$\lambda = \sum_\ell |h_\ell|, \quad \bar{H} = H/\lambda, \quad X_\epsilon = \sin(\epsilon \bar{H}), \quad q_n = \sum_{m=0}^{n} c_{n,m} T_m(X_\epsilon)\Phi_0, \quad n = 0, \ldots, 7.$$

There are $n+1$ coefficients at iteration n. Each colored line below holds the Chebyshev order m fixed and follows $c_{n,m}$ as n increases. A line begins at $n = m$, when that coefficient first exists. The vertical axis is $\log_{10}|c_{n,m}|$: a height of 4 means $|c_{n,m}| \approx 10^4$. All plotted coefficients are real and positive.



Why they grow. The Lanczos recurrence and Chebyshev multiplication rule give

$$\beta_{n+1} q_{n+1} = X_\epsilon q_n - \alpha_n q_n - \beta_n q_{n-1}, \qquad xT_0(x) = T_1(x), \quad xT_m(x) = \tfrac{1}{2}\big(T_{m-1}(x) + T_{m+1}(x)\big)\ (m \geq 1).$$

If $d_{n,m}$ is the coefficient of T_m in $X_\epsilon q_n$, then

$$c_{n+1,m} = \frac{d_{n,m} - \alpha_n c_{n,m} - \beta_n c_{n-1,m}}{\beta_{n+1}}.$$

The small β_{n+1} values repeatedly divide the coefficients. At $\epsilon = 0.5$, $\alpha_0 = -0.14926558$ and $\beta_1 = 0.03029585$, so

$$c_{1,0} = -\alpha_0/\beta_1 = 4.92693, \qquad c_{1,1} = 1/\beta_1 = 33.00782.$$

The code evaluates each state with the Clenshaw recurrence, normalizes it, and applies the same normalization factor to its coefficient row.

Measured growth and its interpretation

Largest coefficient at each iteration. These are $\max_m |c_{n,m}|$; the complete values are in `output/ocqkudsin/chebyshev_coefficients.csv`.

n	$\epsilon = 0.5$	$\epsilon = 0.8$	$\epsilon = 1.2$
0	1	1	1
1	33.0	20.9	14.3
2	419	174	86.7
3	2.20×10^4	5.91×10^3	2.10×10^3
4	4.67×10^5	8.23×10^4	2.59×10^4
5	3.97×10^7	4.71×10^6	8.93×10^5
6	2.26×10^9	1.74×10^8	2.71×10^7
7	2.34×10^{11}	1.19×10^{10}	1.16×10^9

At $n = 7$, the largest entry is $c_{7,1}$, not the newly introduced highest-order coefficient $c_{7,7}$. For these three settings, every fixed-m magnitude increases through $n = 7$; this is an observed pattern, not a general theorem.

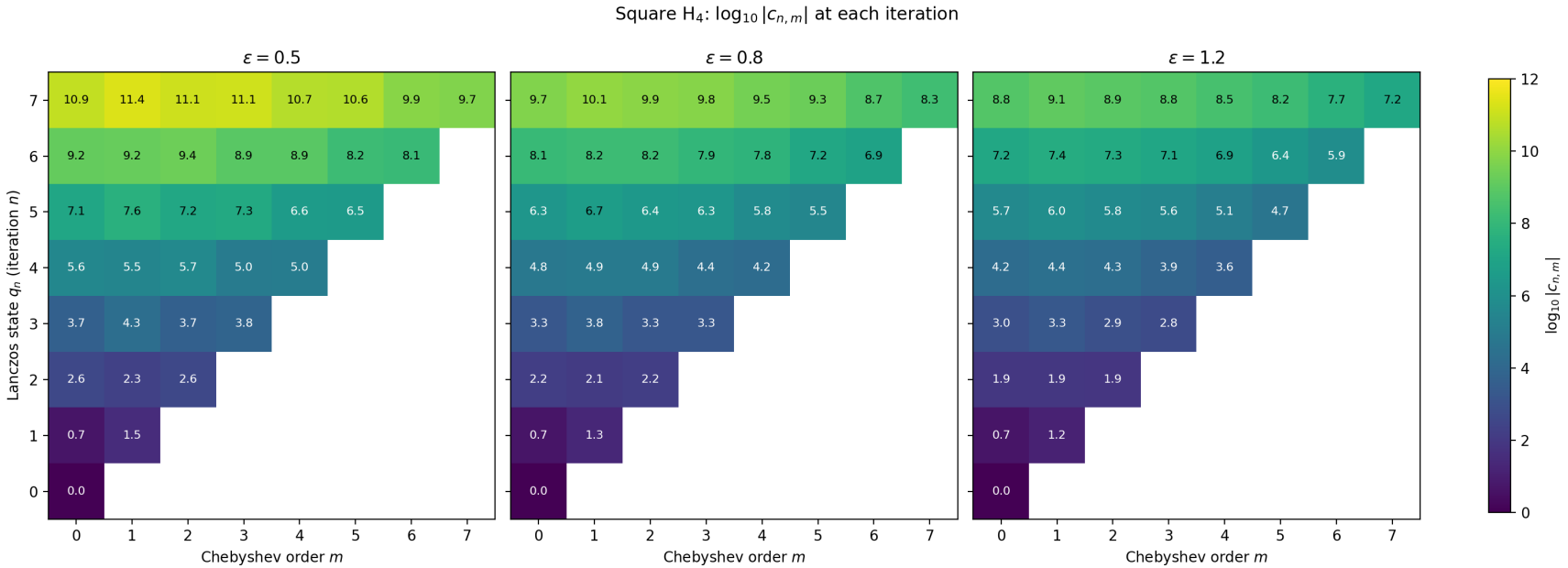
Why a normalized state can have huge coefficients. The Chebyshev vectors $v_m = T_m(X_\epsilon)\Phi_0$ are not orthogonal. With $G_{mk} = \langle v_m | v_k \rangle$, normalization means

$$\|q_n\|^2 = c_n^\dagger G c_n = 1,$$

which does not require the individual $c_{n,m}$ to be small. Even at the first step, the two terms cancel along Φ_0:

$$\langle \Phi_0 | q_1 \rangle = 4.92693 + 33.00782\,\alpha_0 \approx 0.$$

For $\epsilon = 0.5$, the sum of the norms of the eight separate terms in q_7 is about 3.6×10^{11}, while their combined state has norm 1. The matrix with the eight v_m as columns has 2-norm condition number about 6.5×10^{11}. This is *not* the Lanczos-state overlap matrix S; its reported condition number at dimension eight is about 1.000002.
The smaller ϵ here gives smaller early β values and larger Chebyshev coefficients. That ordering holds for these three H_4 runs; coefficient size alone does not establish energy convergence or loss of Lanczos orthogonality.



Heatmap reading: each colored cell is $\log_{10} |c_{n,m}|$; blank cells have $m > n$, so that coefficient does not exist at that iteration.